

Learner Guide: The National Information Ecosystem

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Module purpose

Participants learn to see official statistics as a national information ecosystem rather than as the output of one national statistical office. They identify the institutions, data assets, technologies, flows, users, rules, capabilities and risks that determine whether information is fit for public decisions. They then use that map to locate realistic AI opportunities and the dependencies that must be addressed first.

Performance outcome

Given a country scenario or an approved national domain, the participant can produce a scoped ecosystem map with stable identifiers, evidence references, critical flows, user needs, quality and governance risks, coordination gaps and a justified statement of where AI may or may not add value.

Module	A1
Competencies	C1 Statistical purpose and public value; C3 stewardship; C4 quality; C6 governance; C8 communication.
Entry level	No prior AI expertise required; basic familiarity with a statistical organization or sector.
Exit level	L1 guided practitioner, with an optional L2 workplace assignment.
Structured time	6 hours, delivered as one day or two 3-hour sessions.
Practice share	Approximately 65 percent.
Primary evidence	Completed ecosystem map, flow analysis, risk statement and oral defence.
Phase 1 link	Uses the concepts and record types of Phase 1D.

Package contents

Part	Use
Use	Learner edition. Facilitator answers and scoring guidance are intentionally excluded.

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Learning outcomes and evidence

By the end participants can	Evidence
Define the boundary of a national or sector information ecosystem.	Scope statement naming domain, geography, period, users and exclusions.
Distinguish producer, source owner, steward, processor, coordinator, funder, regulator and user roles.	Actor register with role and authority evidence.
Inventory data assets from surveys, censuses, administrative records, research, geospatial and private sources.	Asset register with provenance and quality notes.
Trace one critical information flow from collection or generation to use.	Flow record showing transformations, handoffs, delays and failure points.
Connect user needs to fitness-for-use requirements.	Need-to-asset matrix with unmet or partially met needs.
Identify coordination, capability, quality, legal, security and sustainability risks.	Prioritized gap and risk list.
Locate bounded AI opportunities without assuming AI is the answer.	Opportunity and dependency statement with a proceed, investigate or do not pursue recommendation.

Critical safeguards

- Do not include confidential records, passwords, security details or personal data in the classroom map.
- Label claims as evidenced, reported or unknown. Do not convert informed opinion into fact.
- Map actual decision authority; do not assume the NSO controls every sector source.
- An AI opportunity is not selected until purpose, data access, quality, human responsibility and security are considered.

Core concept the information ecosystem

A national information ecosystem is the connected set of people, institutions, rules, data assets, technologies, processes, funding arrangements and relationships through which information is generated, transformed, governed, disseminated and used. It includes official statistics but is wider than the national statistical office and wider than the formal national statistical system.

Component	Questions to ask
Public purpose and users	Which decisions, rights, services, markets or public debates require information? Who uses it and when?
Actors and authority	Who owns sources, produces statistics, sets methods, coordinates, regulates, finances, supplies technology and uses results?

Component	Questions to ask
Data and information assets	What surveys, censuses, administrative records, research outputs, geospatial sources, private data, classifications and metadata exist?
Flows and processes	How does information move, transform, link, aggregate, validate, revise and reach users?
Quality and fitness	Which quality dimensions matter for the intended use, and where can error or delay enter?
Rules and safeguards	What laws, agreements, access controls, confidentiality rules, standards and professional responsibilities apply?
Technology and capability	Which systems, skills, infrastructure, support and maintenance arrangements enable the work?
Resources and incentives	Who funds recurrent work? What incentives support or obstruct sharing, quality and coordination?
Risks and resilience	Where are single points of failure, unknowns, dependencies, vulnerabilities and fallback arrangements?

Three nested views

View	Boundary	Example
National	All major producers, sources and users across domains.	National development indicators and cross-government data flows.
Domain	One sector and its links to the wider system.	Food and agriculture information ecosystem.
Product or use case	One statistical output or decision chain.	Seasonal crop forecast from field, administrative and satellite sources.

Roles are not the same as institutions

One institution may perform several roles, and one role may be distributed across institutions. Mapping only organization names hides accountability gaps and competing claims.

Role	Accountability question	Agricultural example
Source owner	Who controls the original operational or observed data?	Veterinary service owns animal-health records.
Producer	Who transforms data into a statistical output?	Agricultural statistics unit estimates livestock numbers.
Methodological authority	Who approves concepts, methods,	NSO or designated sector statistical

Role	Accountability question	Agricultural example
	revisions and release?	authority.
Steward	Who maintains metadata, access, quality and change communication?	Administrative registry data steward.
Coordinator	Who sets common standards and resolves duplication?	NSO under statistical legislation or council arrangements.
Technology provider	Who operates platforms, models or infrastructure?	Government digital service, vendor or research partner.
Funder	Who pays development and recurring costs?	Treasury, ministry budget or development programme.
Regulator or oversight body	Who enforces legal, privacy, security or ethical requirements?	Data protection authority or cybersecurity agency.
User or affected group	Who makes decisions or bears consequences?	Farmers, planners, traders, researchers and the public.

Evidence labels

Label	Meaning	Acceptable basis
E Evidenced	Supported by a document, dataset, system record or authorized observation.	Citation or reference ID.
R Reported	Stated by a knowledgeable respondent but not independently confirmed.	Respondent role and date.
U Unknown	Information unavailable, conflicting or outside current scope.	Question and proposed evidence owner.

What counts as an information asset

Type	Examples	Common mapping error
Survey or census	Agricultural census, household survey, business survey, crop-cutting survey.	Listing a publication but not the microdata, frame, instruments or metadata.
Administrative record	Land registration, veterinary services, subsidies, school enrollment, tax records.	Assuming operational coverage equals statistical population coverage.
Research information	Variety trials, experimental stations, longitudinal studies, laboratory results.	Generalizing experimental findings without external-validity evidence.
Geospatial or sensor source	Satellite imagery, weather stations, GPS traces, remote sensors.	Treating model-derived estimates as direct observation.

Type	Examples	Common mapping error
Private-sector source	Market transactions, mobile services, platform or supply-chain data.	Ignoring contract stability, selection and commercial incentives.
Reference asset	Classifications, frames, geographies, code lists, metadata and quality reports.	Treating reference data as secondary when it controls integration.
Statistical product	Indicator, table, register, map, forecast, report or release.	Failing to link output to source assets and users.

Asset record minimum

Each asset receives a stable ID and records owner, purpose, population or coverage, unit, geography, period, frequency, access, format, classifications, identifiers, transformations, quality evidence, confidentiality class, dependencies, known changes, users and fitness decisions. Unknown fields remain visible.

Mapping information flows and needs

Flow element	Record
Trigger	Event or schedule that starts collection, extraction or processing.
Source and destination	Actor and asset IDs at each handoff.
Transformation	Editing, coding, linkage, modeling, aggregation, seasonal adjustment or interpretation.
Control	Validation, approval, access or quality check at the handoff.
Timing	Frequency, deadline, delay and synchronization requirements.
Failure mode	Loss, duplication, inconsistency, bias, late delivery, undocumented change or unauthorized access.
Responsibility	Named role for correction, acceptance and escalation.
Fallback	Manual route, alternative source, previous method or service continuity arrangement.

Need-to-asset test

1. State the user decision, population, geography, reference period and required timing.
2. Translate the decision into quality and interpretability requirements.
3. Identify current assets and flows that could meet the need.
4. Assess whether each asset is fit, conditionally fit or not fit.
5. Record unmet requirements, evidence gaps and consequences.
6. Only then consider process improvement or AI assistance.

Fictional country case Republic of Kintaba

Kintaba is a lower-middle-income country with 24 million people. Agriculture employs a large share of the rural population and is exposed to drought, pests and price volatility. The government needs earlier district crop-production estimates for food-security planning. The following case is fictional and contains deliberate gaps and contradictions.

Actor ID	Actor	Reported role and situation
ACT-01	National Statistics Directorate	Runs the agricultural census every ten years and an annual production survey. Its frame is based on the last census.
ACT-02	Ministry agriculture statistics unit	Publishes seasonal crop estimates from extension reports and a farmer sample. Methods differ by region.
ACT-03	Extension service	Collects monthly crop-condition forms from local officers. Coverage is better near roads.
ACT-04	Land administration agency	Maintains registered parcels, but customary holdings are often absent.
ACT-05	Meteorological service	Operates 82 stations; 19 had substantial missing observations last season.
ACT-06	National food-security secretariat	Combines crop, market, nutrition and rainfall information for cabinet alerts.
ACT-07	Agricultural research institute	Runs crop-variety trials at six research stations and selected farmer fields.
ACT-08	University geospatial laboratory	Produces experimental crop maps from free satellite imagery with external project funding.
ACT-09	Market information service	Collects weekly prices by phone from selected markets.
ACT-10	Mobile network operator	Holds aggregated mobility and mobile-money indicators; no standing data-sharing agreement exists.
ACT-11	Farmer organizations	Report that official estimates arrive late and rarely include small irrigated plots or gender-disaggregated access information.
ACT-12	Government digital service	Hosts ministry systems but does not manage sector metadata or statistical methods.

Case evidence

Evidence ID	Extract
EVD-01	Statistical law assigns national coordination to ACT-01 and allows designated sector producers.
EVD-02	A memorandum requires ACT-02 to send survey microdata and metadata to ACT-01 within 90 days; last year only tables were sent.
EVD-03	ACT-02 reports 4.8 million tonnes of maize; ACT-01 reports 4.2 million for the same season. Definitions and revision status are not shown.
EVD-04	ACT-08 reports 86 percent crop-classification accuracy on 600 labelled fields in three accessible districts.
EVD-05	ACT-07 trial results show two varieties with higher mean yields under controlled water management. Adoption conditions outside trial sites are not assessed.
EVD-06	Food-security alerts are due six weeks before the annual survey estimate is released.
EVD-07	No common identifier links holdings, parcels, extension reports and survey units.
EVD-08	The geospatial project has eight months of funding remaining and one trained analyst.

Exercise 1 Map actors authority and evidence

Time: 45 minutes. Work in groups. Use ACT and EVD identifiers. Produce an actor register and a short authority analysis.

Task	Required response
A	Assign each actor its relevant roles: source owner, producer, methodological authority, steward, coordinator, technology provider, funder, oversight or user. Multiple roles are allowed.
B	For each assignment, label it E, R or U and cite an evidence ID or the case statement.
C	Identify three authority or accountability questions that must be resolved before a national crop-estimation integration project.
D	Name one actor missing from the case and explain why it matters.

Submission template

Actor ID	Role	Evidence label and reference	Authority or coordination issue

Exercise 2 Inventory and assess assets

Time: 45 minutes. Identify at least eight assets, covering four different source types. Do not confuse the producing institution with the asset or the published output.

Asset ID	Asset	Type	Owner	Main quality or fitness issue
AST-01	Annual agricultural production survey	Survey	ACT-01	Frame age and release timing.

Challenge questions

- Which asset is most authoritative for each quantity, and why?
- Which assets are timely but potentially biased or incomplete?
- Which research or geospatial result is being generalized beyond its evidence?
- Which metadata or reference asset is missing?
- Which assets can be combined now, only conditionally, or not yet?

Exercise 3 Trace the crop-estimate flow

Time: 60 minutes. Trace the flow from farm conditions to a cabinet food-security decision. Include at least five actors, three assets and four transformations or handoffs.

Step	Source actor and asset	Transformation or handoff	Control and owner	Failure or delay	Destination
1					
2					
3					
4					
5					
6					

Required conclusion

Identify the two most consequential failure points. For each, state the affected user decision, consequence, present evidence, missing evidence, responsible role and a realistic non-AI improvement.

Exercise 4 User need risk and AI opportunity

Time: 40 minutes. Screen the proposal: “Use generative AI and satellite imagery to produce district crop estimates automatically.” Do not redesign the entire national system. Produce a bounded recommendation.

Screen	Question
Purpose	What exact decision and statistical product are being improved?
Baseline	What current method, timing, cost and error evidence will the proposal be compared with?
Data	Which sources are permitted, representative, sufficiently documented and stable?
Quality	What benchmark, districts, crops, subgroups, seasons and tolerances are required?
Authority	Who owns the estimate, method, review and release decision?
Human oversight	Which qualified person can challenge or override the result?
Security and confidentiality	What information may enter each tool and environment?
Sustainability	What skills, funding, infrastructure, supplier and fallback dependencies exist?
Decision	Proceed to scoping, investigate dependencies first, or do not pursue? Under what conditions?

Individual assessment prompt

In no more than 500 words, define the ecosystem boundary for improving the timeliness of Kintaba's district crop-production estimates. Identify the critical actors, assets and flow; distinguish evidenced, reported and unknown claims; state the leading fitness-for-use problem; and recommend one next action. Your answer must explain why that action comes before or instead of an AI pilot.

Workplace assignment

Within 30 days, map one approved statistical product or domain segment in the participant's institution. The assignment must remain manageable: normally five to fifteen actors, five to twenty assets and one critical flow.

Deliverable	Minimum
Scope and purpose	One page naming user decision, boundary, accountable sponsor and exclusions.
Actor register	Stable IDs, roles, authority reference, contact role and evidence status.
Asset register	Stable IDs, provenance, access, classifications, quality evidence and users.
Critical flow	Source-to-use handoffs, transformations, controls, delays, owner and fallback.
Need and fitness matrix	At least two user needs assessed as fit, conditionally fit or not fit.
Gap and action list	No more than five priority actions with owner, dependency and evidence need.
AI opportunity note	One bounded opportunity or a justified decision that AI is not presently appropriate.
Supervisor confirmation	Confirms safe scope, factual review and intended follow-up.

Follow-up questions

- What did the institutional chart fail to reveal?
- Which important claim remained unknown, and who could resolve it?
- Where did information lose meaning, quality, time or accountability?
- What improvement is possible without AI?
- What dependency must be solved before an AI use case can be screened?